DYNECHO ACOUSTIC CALCULATOR
ISO-ALIGNED ACOUSTIC REPORT

DynEcho Acoustic Calculator

DynEcho Operator Deck · Acoustic performance proposal

CONSULTANT
DynEcho Acoustic Consulting

PROFILE
Consultant issue

ISSUE
DEC-2026-001 · Rev 00

ISSUED TO
Client delivery team

DATE
30 April 2026

PREPARED BY
DynEcho Operator

PRIMARY ANSWER

R'w 49 dB

DynEcho Operator Deck currently reads R'w 49 dB. DynEcho is carrying the stack on the double leaf route with a scoped estimate posture. Concept issue under the consultant issue profile in building prediction context. 3 citations are attached and 6 live warnings remain explicit.

DYNAMIC ROUTE

Double Leaf

Mass Law anchor is active with double leaf porous fill delegate.

VALIDATION

Scoped estimate

Mass Law anchor is active. Double Leaf is the current airborne family read with medium confidence. Solver spread is currently 20 dB across the candidate family set. No exact wall source row is active for this route; keep it as a formula-owned/source-gated scoped estimate rather than a lab or field measurement claim. Read this as a supported wall estimate, not as a measured claim.

REFERENCE BASIS

ISO 717-1 · ISO 16283-1 · ISO 16283-2 · ISO 12354-1

Wall · Building prediction

ISO 717-1 ISO 16283-1 ISO 16283-2 ISO 12354-1

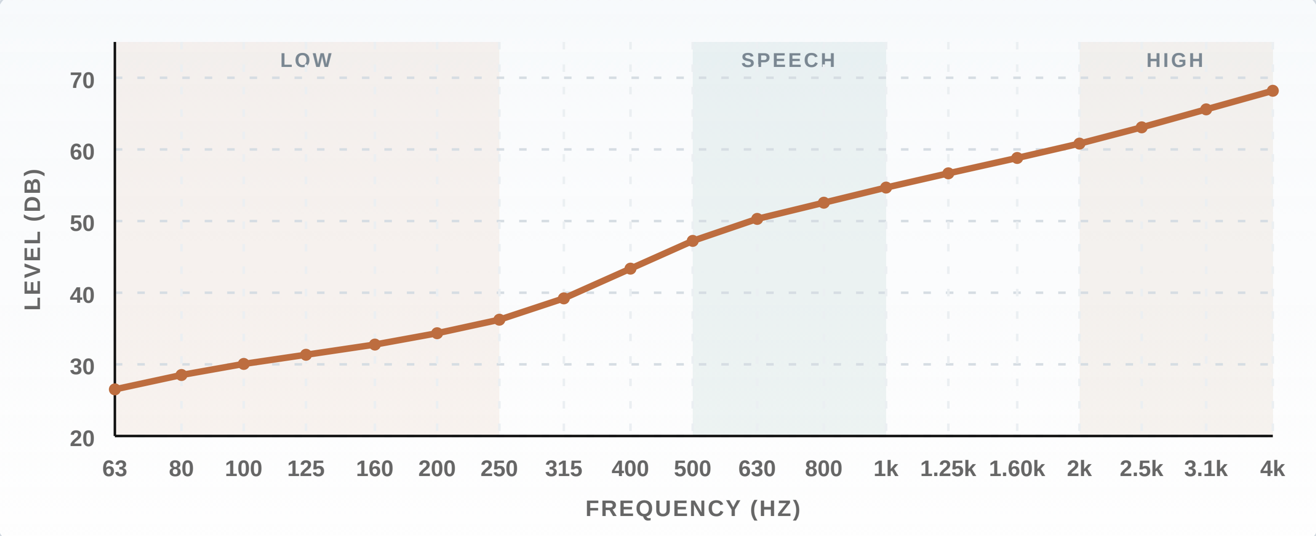
FREQUENCY RESPONSE CURVES

Airborne response curve

63..4000 Hz

Higher is better

Low · 63-250 Hz | Speech · 500-1k Hz | High · 2k-4k Hz



LOW BAND	SPEECH BAND	HIGH BAND
31.3 dB	**47.2 dB**	**60.8 dB**
125 Hz	500 Hz	2k Hz

Final apparent airborne curve

MEASURED / PREDICTED INDICES

INDEX	VALUE	NOTE
R'w	**49 dB**	Building prediction route is active. R'w is being read as the apparent on-site airborne single number from the final apparent curve after the active leakage and flanking overlays.
DnT,w	**50 dB**	Building prediction route is active. DnT,w is being standardized from the same apparent field curve using the current partition area and receiving-room volume.
DnT,A	**48.4 dB**	Building prediction route is active. DnT,A is being carried as DnT,w + C on the standardized field lane.
Dn,w	**50 dB**	Building prediction route is active. Dn,w is being derived from the same apparent field curve using the current partition area only.
Dn,A	**48.5 dB**	Building prediction route is active. Dn,A is being carried as Dn,w + C from the same apparent field curve and partition area.
STC	**49 dB**	ASTM single-number companion from the same airborne curve.

VISIBLE LAYER SCHEDULE

#	MATERIAL	THICKNESS	ROLE
1	Gypsum Board	12.5 mm	Finish
2	MLV Sound Barrier	4 mm	Finish
3	Gypsum Board	12.5 mm	Finish
4	Rock Wool	50 mm	Insulation
5	Rock Wool	50 mm	Insulation
6	Gypsum Board	12.5 mm	Finish
7	Gypsum Board	12.5 mm	Finish
8	Gypsum Plaster	10 mm	Finish
1 additional layer omitted from this short-form layer table; the construction section still uses all 9 solver rows.			

METHOD BASIS

- **Dynamic route**
  Double Leaf · Mass Law anchor is active with double leaf porous fill delegate.

- **Validation posture**
  Scoped estimate · Mass Law anchor is active. Double Leaf is the current airborne family read with medium confidence. Solver spread is currently 20 dB across the candidate family set. No exact wall source row is active for this route; keep it as a formula-owned/source-gated scoped estimate rather than a lab or field measurement claim. Read this as a supported wall estimate, not as a measured claim.

- **Study scope**
  R'w · Wall · Building prediction · Consultant issue

REFERENCE BASIS

- **ISO 717-1 · Weighted airborne ratings**
  Weighted airborne sound insulation rating

- **ISO 16283-1 · Field airborne route**
  Field measurement of airborne sound insulation

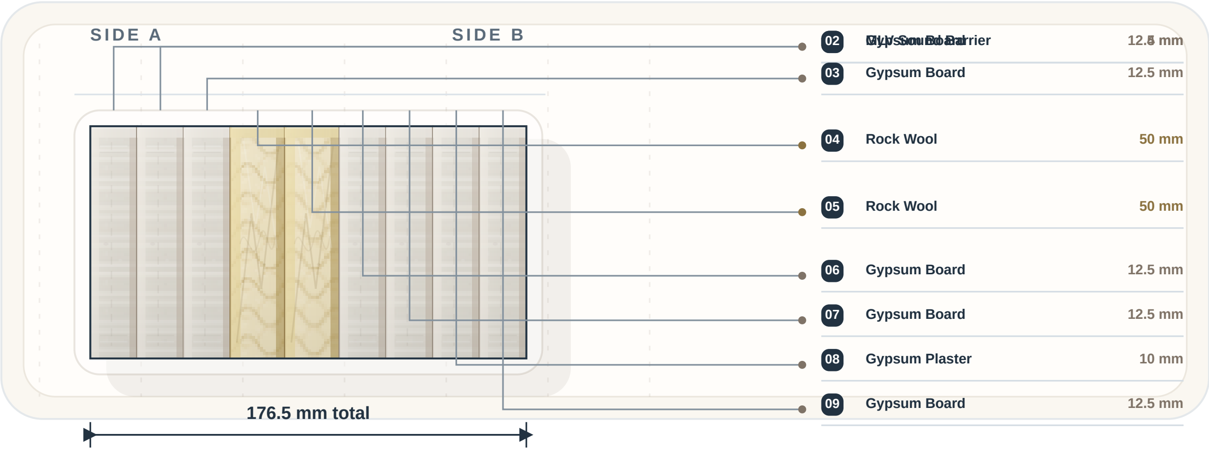
- **ISO 16283-2 · Field impact route**
  Field measurement of impact sound insulation

- **ISO 12354-1 · Prediction route**
  Building acoustics prediction model

OUTPUT COVERAGE REGISTER

METRIC	STATUS	EVIDENCE CLASS	CURRENT STATE
Rw	Unsupported on lane	**Unsupported on route** The active topology does not expose a defensible solver lane for this metric. Keep it visible, but frame it as out of scope on the current route.	**Not ready** Live from the calibrated airborne curve and curated floor-family companions.
R'w	Live now	**Field continuation** This metric is carried through the active field continuation chain from the current lab or apparent curve. It is not being framed as an independent exact source row.	**49 dB** Building prediction route is active. R'w is being read as the apparent on-site airborne single number from the final apparent curve after the active leakage and flanking overlays.
DnT,w	Live now	**Field continuation** This metric is carried through the active field continuation chain from the current lab or apparent curve. It is not being framed as an independent exact source row.	**50 dB** Building prediction route is active. DnT,w is being standardized from the same apparent field curve using the current partition area and receiving-room volume.
DnT,A	Live now	**Field continuation** This metric is carried through the active field continuation chain from the current lab or apparent curve. It is not being framed as an independent exact source row.	**48.4 dB** Building prediction route is active. DnT,A is being carried as DnT,w + C on the standardized field lane.
Dn,w	Live now	**Field continuation** This metric is carried through the active field continuation chain from the current lab or apparent curve. It is not being framed as an independent exact source row.	**50 dB** Building prediction route is active. Dn,w is being derived from the same apparent field curve using the current partition area only.
Dn,A	Live now	**Field continuation** This metric is carried through the active field continuation chain from the current lab or apparent curve. It is not being framed as an independent exact source row.	**48.5 dB** Building prediction route is active. Dn,A is being carried as Dn,w + C from the same apparent field curve and partition area.
STC	Live now	**Companion carry-over** This number is a companion term derived from the primary live lane, not a separate solver route with its own evidence anchor.	**49 dB** ASTM single-number companion from the same airborne curve.
Ctr	Live now	**Companion carry-over** This number is a companion term derived from the primary live lane, not a separate solver route with its own evidence anchor.	**-7 dB** Traffic-noise adaptation term on the airborne lane.

CONSTRUCTION SECTION



TOTAL THICKNESS

176.5 mm total

Side A to Side B in solver order.

LAYER DETAIL

#	MATERIAL	THICKNESS	DENSITY	SURFACE MASS
1	Gypsum Board	12.5 mm	850 kg/m³	10.6 kg/m²
2	MLV Sound Barrier	4 mm	1,900 kg/m³	7.6 kg/m²
3	Gypsum Board	12.5 mm	850 kg/m³	10.6 kg/m²
4	Rock Wool	50 mm	45 kg/m³	2.3 kg/m²
5	Rock Wool	50 mm	45 kg/m³	2.3 kg/m²
6	Gypsum Board	12.5 mm	850 kg/m³	10.6 kg/m²
7	Gypsum Board	12.5 mm	850 kg/m³	10.6 kg/m²
8	Gypsum Plaster	10 mm	1,000 kg/m³	10 kg/m²
1 additional layer omitted from this short-form detail table; the construction section still uses all 9 solver rows.				

SOURCE NOTES

- **Field airborne provenance: Room-standardized apparent derivation**
  Building prediction route is active. DnT outputs are being standardized from the final apparent curve after the active leakage and flanking overlays using partition area and receiving-room volume. Building prediction is the active route label.

- **Field airborne derivation lines**
  Airborne field provenance: Room-standardized apparent derivation. Building prediction route is active. DnT outputs are being standardized from the final apparent curve after the active leakage and flanking overlays using partition area and receiving-room volume. R'w provenance: apparent on-site airborne single number from the final field curve Dn,w provenance: area-normalized apparent derivation via 10log10(A0/S) Dn,A provenance: Dn,w + C on the area-normalized apparent lane DnT,w provenance: room-standardized apparent derivation via 10log10(0.32V/S) DnT,A provenance: DnT,w + C on the standardized apparent lane

CONSULTANT NOTE AND ASSUMPTIONS

- **Consultant note**
  Record assumptions, flanking risks, and report caveats here.

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WARNINGS AND ISSUE GUARDRAILS

- **Issue note**
  Airborne estimate is using the Dynamic Topology path instead of the screening seed.
- **Issue note**
  The selected airborne lane is local to this repo; higher-order family solvers and broader comparison envelopes are still being expanded.
- **Issue note**
  Lightweight assemblies still need broader local family coverage before the selected airborne lane can be treated as the highest-confidence path.
- **Issue note**
  Airborne field-side overlay active. The current building prediction context is carrying a conservative flanking penalty of 1.8 dB.
- **Issue note**
  Unsupported target outputs: Rw. These outputs were requested but not calculated.